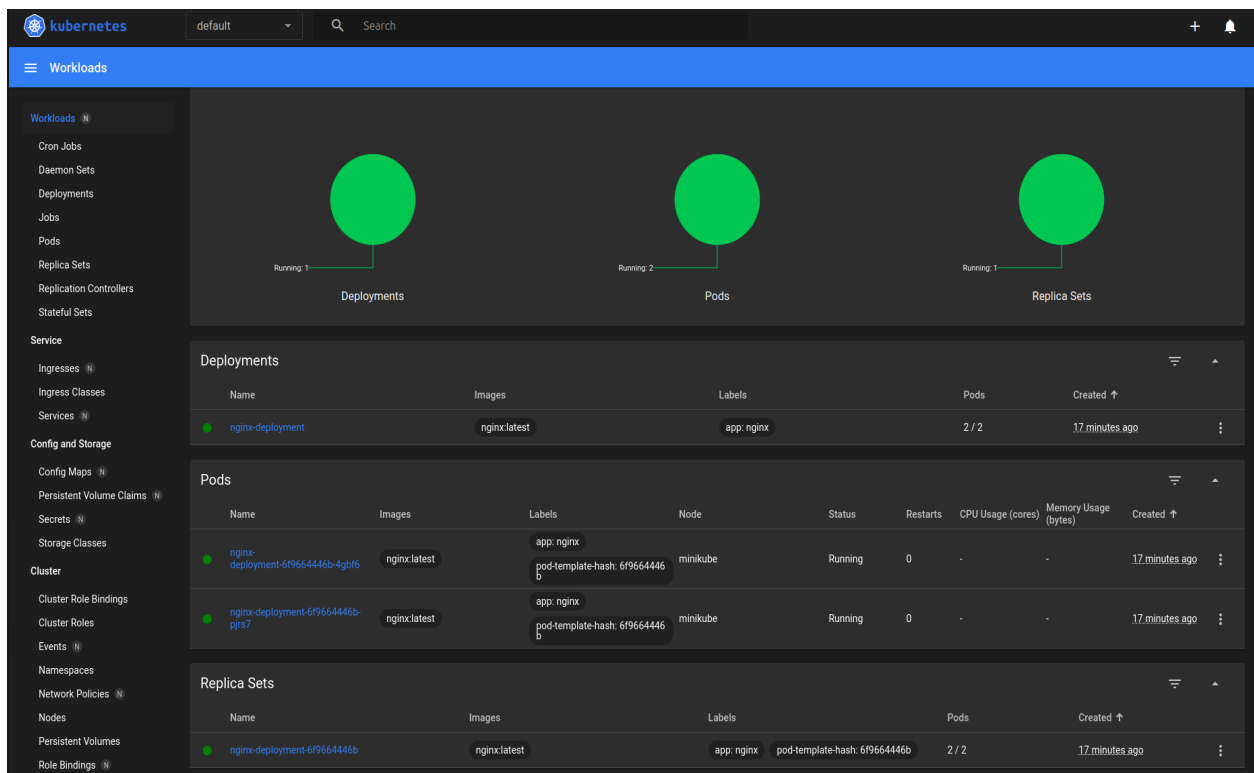


Program 10: Demonstrate k8s dashboard and CLI for Cluster Monitoring

Manifest Files:



Deployment and Pods Description

```
1RV24MC096_SAURABHR@sentient:~$ cd Docker_MCA/
1RV24MC096_SAURABHR@sentient:~/Docker_MCA$ mkdir program10
1RV24MC096_SAURABHR@sentient:~/Docker_MCA$ cd program10
1RV24MC096_SAURABHR@sentient:~/Docker_MCA/program10$ minikube start --cpus=2 --memory=4096
🐹 minikube v1.37.0 on Ubuntu 24.04
🔗 Using the docker driver based on existing profile
! You cannot change the memory size for an existing minikube cluster. Please first delete the cluster.
🔥 Starting "minikube" primary control-plane node in "minikube" cluster
📦 Pulling base image v0.0.48 ...
🔗 Updating the running docker "minikube" container ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
   ▪ Using image docker.io/kubernetes/dashboard:v2.7.0
   ▪ Using image docker.io/kubernetes/metrics-scraper:v1.0.8
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

🌟 Enabled addons: storage-provisioner, default-storageclass, dashboard
🏁 Done! kubectrl is now configured to use "minikube" cluster and "default" namespace by default
```

```
1RV24MC096_SAURABHR@sentient:~/Docker_MCA/program10$ nano deployment.yaml
1RV24MC096_SAURABHR@sentient:~/Docker_MCA/program10$ nano service.yaml
1RV24MC096_SAURABHR@sentient:~/Docker_MCA/program10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment created
1RV24MC096_SAURABHR@sentient:~/Docker_MCA/program10$ kubectl apply -f service.yaml
service/nginx-service created
1RV24MC096_SAURABHR@sentient:~/Docker_MCA/program10$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-4gbf6   1/1     Running   0           34s
nginx-deployment-6f9664446b-pjrs7   1/1     Running   0           34s
1RV24MC096_SAURABHR@sentient:~/Docker_MCA/program10$ kubectl get svc
NAME      TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes  ClusterIP   10.96.0.1     <none>        443/TCP          14d
nginx-service NodePort    10.103.209.70 <none>        80:31370/TCP     32s
1RV24MC096_SAURABHR@sentient:~/Docker_MCA/program10$ kubectl describe pods nginx-deployment-6f9664446b-4gbf6
Name:          nginx-deployment-6f9664446b-4gbf6
Namespace:     default
Priority:       0
Service Account: default
Node:          minikube/192.168.49.2
Start Time:    Wed, 31 Dec 2025 20:23:34 +0530
Labels:        app=nginx
               pod-template-hash=6f9664446b
Annotations:   <none>
Status:        Running
IP:            10.244.0.57
IPs:           10.244.0.57
Controlled By: ReplicaSet/nginx-deployment-6f9664446b
Containers:
  nginx:
    Container ID:  docker://2261bcbdda51b6c71698705b68503522382837dd258d042f72b6097e8f168fa2
    Image:         nginx:latest
    Image ID:      docker-pullable://nginx@sha256:ca871a86d45a3ec6864dc45f014b11fe626145569ef0e74deaafc95a3b15b430
    Port:         80/TCP
    Host Port:     0/TCP
    State:        Running
      Started:    Wed, 31 Dec 2025 20:23:42 +0530
    Ready:        True
    Restart Count: 0
    Environment:  <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-cqvms (ro)
Conditions:
  Type              Status
  PodReadyToStartContainers  True
  Initialized         True
  Ready               True
  ContainersReady     True
```

Events:

Type	Reason	Age	From	Message
Normal	Scheduled	101s	default-scheduler	Successfully assigned default/nginx-deployment-6f9664446b-4gbf6 to minikube
Normal	Pulling	100s	kubelet	Pulling image "nginx:latest"
Normal	Pulled	93s	kubelet	Successfully pulled image "nginx:latest" in 7.849s (7.849s including waiting). Image size: 151914258 bytes.
Normal	Created	93s	kubelet	Created container: nginx
Normal	Started	93s	kubelet	Started container nginx

1RV24MC096_SAURABHR@sentient:~/Docker_MCA/program10\$ minikube service nginx-service

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:31370

🔗 Opening service default/nginx-service in default browser...

1RV24MC096_SAURABHR@sentient:~/Docker_MCA/program10\$ minikube dashboard

🐼 Verifying dashboard health ...

🔗 Launching proxy ...

🐼 Verifying proxy health ...

🔗 Opening http://127.0.0.1:42859/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ in your default browser...

ln: failed to create symbolic link '/home/1RV24MC096_SAURABHR/snap/firefox/7477/.config/dconf/user': File exists

Gtk-Message: 20:21:20.701: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.